lessons 1 to 5, intro & foundation

lesson 6 Universal vs. existence

Prop 1 The sum of (any) 2 even integers is even

universal $\longleftrightarrow$ conditional

if x and y are even, then x+y is even

Proof Let x and y be (arbitrary) even integers, then $x = 2n$ and $y = 2m$ for some $n, m \in \mathbb{Z}$. So, $x+y = 2n+2m$
$= 2(n+m)$.
Since $n+m$ is an integer, x+y is even by def (of even)

Prop 2 Let a, b, c, d be integers. If $a|b$ and $c|d$, then $ac|bd$.
(universal conditional)

If $a|b$ and $c|d$, then $b = an$ and $d = cm$ for some $n, m \in \mathbb{Z}$. So, $bd = (an)(cm)$
$= (ac)(nm)$.
Since nm is an integer, we have $ac|bd$ by def (of "|")

Universal statements can't be proved

with a few examples $\underset{\text{even}}{2}+\underset{\text{even}}{4}=\underset{\text{even}}{6}$ X

Use a thm to prove another thm (along w/def's)

<u>Divison Algorithm</u> (DA) (divide a by b)

unique

$a\in\mathbb{Z}$, $b\in\mathbb{N}$, $\exists!$ $q,r\in\mathbb{Z}$

s.t. $a=bq+r$, $0\le r<b$ (proof: lesson 10)

$(9=4(2)+1)$

<u>Thm 3</u> An integer is even or odd but not both.

even: $x=2n$...

odd: $x=2m+1$

<u>Proof</u> let $x\in\mathbb{Z}$. If we divide x by 2, by the DA, $x=2q+r$ for <u>unique integers q,r</u>, s.t. $0\le r<2$.

So, $x=2q$ or $x=2q+1$ but not both. Hence x is even or odd but not both.

Universal: may consider <u>diff cases</u>

<u>Prop 4</u> If x and y (integers) have the same parity (both odd or both even), then $x+y$ is even. 2 cases

Proof ① If x and y are even, then
$x = 2n$, $y = 2m$ ---- (prop 1)

② If x and y are odd, then
$x = 2n+1$ and $y = 2m+1$
$x+y = (2n+1) + (2m+1)$
$= 2(n+m+1)$

$x+y$ is even

Prop 5: If x and y have different parities, then $x+y$ is odd.
(2 case : ① odd + even ② even + odd)
the same : commutative

Proof: Without loss of generality (WLOG), let x be even and y be odd. Then $x = 2n$, $y = 2m+1$
$x+y = 2n + (2m+1)$
$= 2(n+m)+1$
odd

(not for Prop4)

Existence
→ need to show an example

There is $x \in \mathbb{N}$ s.t. $2^x < x^2$

Proof: Let $x = 3$, then

$$2^3 = 8 < 3^2 = 9$$

For every integer x, there is an integer y s.t. $x + y = 0$ (additive inverse)

Proof Let x be an integer, then $y = -x$ is an integer and

$$x + \underbrace{(-x)}_{y} = 0$$

unique

$$\forall x \in \mathbb{Z}, \exists ! y \in \mathbb{Z}, x + y = 0$$

Proof: 1st, existence:

Next, uniqueness:

Suppose $x + y = 0$ and $x + z = 0$, then $\cancel{x} + y = \cancel{x} + z \rightarrow \underbrace{y = z}_{\text{unique}}$.

Disproof

• Disprove universal

$\forall x \in A,\ s(x)$

$\longrightarrow$ give a counterexample x:

s.t. $s(x)$ is false

For all $x \in \mathbb{R}$, $x^2 \geq x$. counterex

Disprove with $x = 0.5$:

$0.5^2 = 0.25 < 0.5$

• Disprove existence

$\exists x \in A,\ s(x)$

Show that for all $x \in A$,

$s(x)$ is false

There is a real # x s.t.

$x^2 + 2x + 2 = 0$

Need to show for all $x \in \mathbb{R}$,

$x^2 + 2x + 2 \neq 0$

$$x^2 + 2x + 2 = \underbrace{(x+1)^2}_{x^2+2x+1} + 1$$

Completing the square

$$\geq \underbrace{0 + 1}_{1}$$

$$x^2+2x+2 \geq 1$$

'' $\neq 0$